



Network Topology

Network Types · Segmentation · Defensible Architecture · Diagram Analysis

420-D01-AB — Cybersecurity Fundamentals

Cohort CYB-01 · Days 03 & 04 · 4 hrs lecture + 4 hrs Lab 02

From the Industry to the Classroom — Building the IT's Future, Today and Together!

The block, at a glance

- ▶ **Lecture 1 (2 h)** — Network types · Segmentation · Defensible architecture
- ▶ **Lab 02 Part 1 (2 h)** — Map architecture · Identify vulnerabilities
- ▶ **Lecture 2 (2 h)** — Analyse a sample network diagram
- ▶ **Lab 02 Part 2 (2 h)** — Document & validate diagrams

KEY TAKEAWAY

Day 03-04 produces two chapters of your Course Procedure Manual (CPM Ch. 2 + Ch. 3).

How Lab 02 fits among the six course labs

- ▶ **Lab 01 (Day 02)** — Cybersecurity Fundamentals — vocabulary for cyber risk
- ▶ **Lab 02 (Days 03–04)** — Map architecture & identify vulnerabilities — we are here
- ▶ **Lab 03 (Day 06)** — Wireshark · SNMP · Splunk on Windows + Linux
- ▶ **Lab 04 (Day 09)** — Traffic and log analysis — IDS alerts · rogue access
- ▶ **Lab 05 (Days 10–11)** — Live threat detection · attack simulation
- ▶ **Lab 06 (Days 12–13)** — Scripted analysis · cloud monitoring

KEY TAKEAWAY

Everything from Day 03 onward references the diagrams you produce in Lab 02.

What you walk out with at the end of Day 04

- ▶ A mental model of why a flat /24 is the default Christmas present to ransomware operators
- ▶ The ability to read a sample diagram and spot the three things wrong with it
- ▶ A scripted, repeatable process for mapping a network you have authority to touch
- ▶ A scripted, repeatable process for documenting what you mapped, readably
- ▶ Two CPM chapters drafted and version-controlled

KEY TAKEAWAY

The competency is the eye, not the brand of the tool.

Lecture 1

Network Types · Segmentation · Defensible Architecture · 2 hours

PAN · LAN · CAN · MAN

PAN — Personal Area Net.

body-width range

- ▶ Bluetooth · smart-watch · phone
- ▶ Pairing / BLE sniffing

LAN — Local Area Network

one physical site

- ▶ The office floor · the dorm
- ▶ ARP poisoning · rogue DHCP

CAN — Campus Area Net.

many sites, one campus

- ▶ University · corporate campus
- ▶ LAN risks × N + propped doors

MAN — Metropolitan Area

city scale

- ▶ Carrier or municipality
- ▶ BGP leaks · cable cuts

KEY TAKEAWAY

The classification chooses the toolbox. A firewall rule that fits a PAN is malpractice in a SAN.

WAN · SAN · VPN — and the trap students fall into

WAN — Wide Area Network

multi-city / multi-country

- ▶ MPLS · SD-WAN · site-to-site VPN
- ▶ Transit compromise · tunnel mis-key

SAN — Storage Area Net.

dedicated block-storage

- ▶ Fibre Channel · iSCSI
- ▶ Compromise = the DISKS

VPN — Virtual Private Net.

a tunnel, not a place

- ▶ Connects two of any of the above
- ▶ Trust model AFTER entry matters

VLAN — Virtual LAN

NOT a network type

- ▶ Segmentation inside a LAN
- ▶ See §3 — Segmentation

KEY TAKEAWAY

VLAN is not in the network-type list. Don't conflate segmentation with topography.

Bus · Ring · Star — failure-mode order, not chronological

- ▶ **Bus** — one cable cut anywhere → entire network down · every host sniffs
- ▶ **Ring** — one node down → ring broken (unless dual-counter-rotating)
- ▶ **Star** — centre switch dies → office dies. This is 99 % of office LANs.

KEY TAKEAWAY

The shape decides the failure mode. The 'type' sells the cake — the shape decides who eats first.

Mesh · Tree · Hybrid

- ▶ **Mesh** — survives almost any single failure · cost is wiring + complexity
- ▶ **Tree** — hierarchy of stars — cut high, lose more
- ▶ **Hybrid** — the honest label for any real production network

KEY TAKEAWAY

Every real network is a hybrid. Calling it anything else is a lie of convenience.

Three segmentation primitives

- ▶ **Subnetting** — split an IP range · routers can drop packets — that is the security
- ▶ **VLAN (Virtual LAN)** — split a switch into logical Layer-2 broadcast domains
- ▶ **Microsegmentation** — firewall rules at the workload level — Zero Trust delivers this

KEY TAKEAWAY

A flat /24 is the default Christmas present to ransomware operators.

Median lateral-movement time : minutes, not days

- ▶ Annual reports from CISA and the Canadian Centre for Cyber Security confirm this every year
- ▶ In a flat network, the attacker is at the crown jewels before the SOC sees the first alert
- ▶ Hamilton ON accounting firm (2024) — phishing → 18 minutes → 5 hosts encrypted
- ▶ All five hosts on 192.168.1.0/24 · one broadcast domain · no internal firewalling
- ▶ Cheap fix : four VLANs + one inter-VLAN ACL with three explicit permits

KEY TAKEAWAY

The bullet you fired was a /24, and that is on you.

The seven layers — Layers 1–4

- ▶ **Layer 1 — Policies & Awareness** — acceptable-use · phishing training · the human layer
- ▶ **Layer 2 — Physical** — locks · doors · cameras · badge readers · mantraps
- ▶ **Layer 3 — Perimeter** — the edge · firewall · IPS (Intrusion Prevention System)
- ▶ **Layer 4 — Internal Network** — LAN-side · VLANs · NAC · 802.1X

KEY TAKEAWAY

Every control fails. Stack them so the next one catches what the previous one missed.

The seven layers — Layers 5–7

- ▶ **Layer 5 — Host** — EDR · patching · antivirus · disk encryption
- ▶ **Layer 6 — Application** — WAF · input validation · authentication boundaries
- ▶ **Layer 7 — Data** — encryption at rest · DLP (Data Loss Prevention) · key management

KEY TAKEAWAY

DiD is what you draw when you want to explain the architecture to the CEO.

Never trust, always verify — NIST SP 800-207

- ▶ Inside-LAN is NOT a credential
- ▶ Login this morning is NOT ongoing authentication
- ▶ Managed laptop is NOT a green light without a posture check
- ▶ Every access request is evaluated against current policy + identity + device posture
- ▶ Zero Trust does NOT replace Defense in Depth — it strengthens the inner-network layer

KEY TAKEAWAY

The internal network no longer provides implicit trust. Verify per-request.

The three load-bearing components of Zero Trust

Policy Decision Point

the brain (PDP)

- ▶ Consumes identity + device + context
- ▶ Returns yes / no per request

Policy Enforcement Pt.

the gate (PEP)

- ▶ Does what the PDP says
- ▶ Lives at app or network boundary

Signals

the inputs

- ▶ AD / SSO identity
- ▶ EDR posture · time · risk

KEY TAKEAWAY

Without good signals, the PDP is just guessing. The signal pipeline is the work.

The Defensible Architecture Checklist — Q1–5

- ▶ 1. What TYPE of network is this ? (LAN / WAN / MAN / SAN / ...)
- ▶ 2. What TOPOLOGY is it actually ? (star, tree, hybrid — never trust the label)
- ▶ 3. How many BROADCAST DOMAINS ? (one is almost always wrong)
- ▶ 4. Where is the boundary between USER and SERVER traffic ?
- ▶ 5. Where is the boundary between IT and OT (Operational Technology), if any ?

The Defensible Architecture Checklist — Q6–10

- ▶ 6. Is ADMINISTRATIVE ACCESS on its own segment ?
- ▶ 7. Is GUEST ACCESS physically and logically separated ?
- ▶ 8. Is there a SINGLE POINT OF FAILURE — physical, logical, or human ?
- ▶ 9. Which DiD LAYERS are present, and which are not ?
- ▶ 10. Are any ZERO TRUST signals actually being collected and acted on ?

KEY TAKEAWAY

If you cannot answer any one of these about the network in front of you — it is undocumented.

Lab 02 — Map & Identify

Hands-on · 2 hours in class · Day 03 (Parts A–D)

Mapping the architecture · identifying vulnerabilities

- ▶ **Part A — Environment** — 20 min · router + 2 switches + 3 hosts on 2 segments
- ▶ **Part B — Three maps** — 50 min · PHYSICAL · LOGICAL · TRUST
- ▶ **Part C — Vulnerabilities** — 40 min · walk the checklist · build Risk Register
- ▶ **Part D — Report scaffold** — 10 min + home · 6 of the final 10 A4 pages · PDF/A

KEY TAKEAWAY

Three maps. Every real network has three. Most networks ship one.

Three maps · three questions answered

Physical map

if a cable comes loose, what stops working ?

- ▶ Devices · interfaces · ports
- ▶ Cable types · virtual links

Logical map

if a packet leaves A for B, what path ?

- ▶ Subnets and masks
- ▶ Default gateways · NAT boundaries

Trust map

if A is compromised, what can A reach ?

- ▶ Allowed protocols / ports
- ▶ Auth bnds · data classification

KEY TAKEAWAY

The trust map is the security map — the one most networks never produce.

The Architectural Risk Register — column shape

- ▶ **# — finding ID** — AR-001, AR-002, ...
- ▶ **Finding** — one-line description of the issue
- ▶ **Where** — subnet · host · diagram region
- ▶ **Severity** — C / H / M / L / I — CVSS v3.1 qualitative bands
- ▶ **Remediation** — implementable, one line — 'buy a different firewall' does NOT count
- ▶ **Effort + Owner** — S / M / L · which role owns the fix

KEY TAKEAWAY

Minimum 6 rows for Part 1 · raise to 10 rows minimum for the Part 2 final.

Submission details — read these once, then do not ask

- ▶ **Format** — PDF/A only · one file for both halves
- ▶ **Length** — max 10 A4 pages total — screenshots included
- ▶ **Channel** — OmniVox LEA Dropbox
- ▶ **Cut-off** — Sunday, 24 May 2026 · 23 h 59 EST (UTC-5)
- ▶ **Late penalty** — -10 % per calendar day
- ▶ **Filename** — LASTNAME_FIRSTNAME_CYB01_Lab02.pdf

KEY TAKEAWAY

Late penalty is automatic. There is no extension for forgetting OmniVox.

Lecture 2

Analyse a Sample Network Diagram · 2 hours

The five elements every good diagram contains

- ▶ **1 — Title block** — what · version · date · author. Otherwise it is folklore.
- ▶ **2 — Legend** — every line style · shape · colour. Otherwise readers invent their own.
- ▶ **3 — Addressing** — IP per subnet · gateway · exit interface. Otherwise decorative.
- ▶ **4 — Trust zones** — labelled regions : Internet · DMZ · Internal · Restricted
- ▶ **5 — Scope note** — what is NOT on this diagram — because none contains everything

KEY TAKEAWAY

Without the scope note, every reader trusts what is drawn as if it were complete.

One source of truth · three derived diagrams

Physical

cables & ports

- ▶ What is wired to what
- ▶ Lose the cable, lose the link

Logical

subnets & routes

- ▶ What speaks what at Layer 3
- ▶ Lose the route, lose the path

Trust

allowed flows

- ▶ What may talk to what at L4+
- ▶ Compromise A → A reaches what ?

KEY TAKEAWAY

Maintain one source of truth, generate three views. Anything else drifts within weeks.

Ten-second triage — every diagram, every time

- ▶ **1 — Where is the Internet ?** — top, bottom, corner ? It must be visible.
- ▶ **2 — Where is the boundary firewall ?** — between Internet and what ?
- ▶ **3 — Where is the DMZ ?** — is there one ? If not, where are the public services ?
- ▶ **4 — Where is administrative access ?** — same VLAN as users (bad) or separate (good) ?
- ▶ **5 — Where is the data ?** — flagged separately, or mixed in with compute ?

KEY TAKEAWAY

If you spend more than 10 seconds finding the Internet, the diagram is failing you.

The next thirty seconds — what to look for

- ▶ Count the BROADCAST DOMAINS — one is almost always wrong
- ▶ Look for the words 'TRUST' and 'UNTRUST' — they should appear
- ▶ Look for REDUNDANCY — single line you can erase that disconnects something critical ?
- ▶ Look for the CLEANER — where does an unsupervised person have access to live jacks ?
- ▶ Look for the PRINTER — most-forgotten persistent foothold in any enterprise

KEY TAKEAWAY

The cleaner and the printer are senior pen-testers' first two questions.

The twelve architectural smells — #1–6

- ▶ 1. One big subnet — no segmentation, lateral movement is free
- ▶ 2. 'Firewall' with no zone labels — it is a router with a sticker
- ▶ 3. Public-facing service with no DMZ — compromised service = inside the LAN
- ▶ 4. Admin interface on user VLAN — admin compromise = user compromise
- ▶ 5. Printers on every VLAN — forgotten persistent foothold
- ▶ 6. IoT devices on user network — TVs · cameras · sensors — unpatchable

The twelve architectural smells — #7–12

- ▶ 7. Wireless on the same VLAN as wired — Wi-Fi compromise = LAN compromise
- ▶ 8. One Internet circuit, no failover — single line, total outage
- ▶ 9. No backup network indicated — backups in-band are a ransomware target
- ▶ 10. No out-of-band management — locked out when most needed
- ▶ 11. 'Cloud' cloud with no detail — trust without verification
- ▶ 12. A diagram older than the team viewing it — folklore, not architecture

KEY TAKEAWAY

A good first audit finds 7 of 12. A great audit finds the human stories behind each.

NorthernEdge Analytics Inc. — your audit target

- ▶ Fictional 40-person data-analytics consultancy · Toronto · one office
- ▶ **northernedge_inventory.csv** — 65 rows · every device on the network
- ▶ **northernedge_network.yaml** — source-of-truth YAML for the diagram generator
- ▶ **09_northernedge_master.svg** — the 'official' as-handed diagram
- ▶ Your job (Lab 02 cont.) — run the four scripts · find what they missed · validate

KEY TAKEAWAY

Scripts catch the obvious. You catch the rest. The graded section is the human one.

Lab 02 cont.

Document & Validate · 2 h in class + 2 h home · Parts E–G

Documenting & validating diagrams

- ▶ **Part E — Run the four scripts** — 40 min · completeness · views · score · smells
- ▶ **Part F — Compile the report** — 40 min · 10 A4 pages total · Part 1 + Part 2 combined
- ▶ **Part G — Validate** — 20 min + home · diagram ↔ inventory ↔ reality

KEY TAKEAWAY

Validation is the step nobody does. We do it explicitly.

What you run, in order, on NorthernEdge

05 — Completeness

Python + pydot

- ▶ Verifies the 5 anatomy elements
- ▶ Output : diagram_check.json

06 — Three views

Python + PyYAML

- ▶ 1 YAML → 3 .dot files
- ▶ Render with Graphviz dot -Tpng

07 — Trust-Boundary Score

PowerShell

- ▶ Single-number KPI from CSV
- ▶ Output : trust_score.json

08 — Smell scanner

Bash + awk

- ▶ Auto-detects 5 of 12 smells
- ▶ Output : Markdown table

KEY TAKEAWAY

Four scripts · four artefacts. Paste each into the final report.

A single-number KPI for the board meeting

- ▶ **Formula** — $100 - 10 \times (\text{bad adjacencies}) \cdot \text{floor } 0$
- ▶ **Bad adjacency** — two devices on same subnet but tagged different trust zones
- ▶ **Grade bands** — $A \geq 90 \cdot B \geq 75 \cdot C \geq 60 \cdot F < 60$
- ▶ **Track YoY** — score should rise after each remediation (Labs 03 → 06)
- ▶ **Why it matters** — single-number KPIs are how security work gets funded

KEY TAKEAWAY

A manager who cannot derive the score on a napkin will not trust it. Keep it simple.

Final report structure — sections 1–7

- ▶ 1. Cover page · executive summary
- ▶ 2. The three maps from Part B (physical · logical · trust)
- ▶ 3. Defensible Architecture Checklist with your answers
- ▶ 4. Architectural Risk Register — 10 rows minimum
- ▶ 5. Completeness JSON (Part E.1) — code block
- ▶ 6. The three derived views (Part E.2) — 3 embedded PNGs + 1 paragraph each
- ▶ 7. Trust-Boundary Score (Part E.3) — score + grade + interpretation

Final report structure — sections 8–12

- ▶ 8. Smell scan (Part E.4) — Markdown table
- ▶ **9. THREE HUMAN-ONLY FINDINGS** — ← graded strictest · worth 18 / 100
- ▶ 10. Reflection — one paragraph : what changed between draft and final
- ▶ 11. Appendix A — raw command outputs (Parts A + E artefacts)
- ▶ 12. Appendix B — every .dot and .yaml source file, pasted as code blocks

KEY TAKEAWAY

Section 9 decides whether you watched the lecture or understood it.

Walk diagram ↔ inventory ↔ reality, document every divergence

- ▶ Is every device on the CSV also on the diagram ?
- ▶ Is every device on the diagram also in the CSV ?
- ▶ When they disagree — which is right ? Trick : neither, unless you check the rack.
- ▶ Do the SUBNETS on the diagram match the subnets in the CSV ?
- ▶ Do the TRUST ZONES on the diagram match those in the CSV ?
- ▶ Instructor has baked AT LEAST FOUR deliberate divergences — find them.

KEY TAKEAWAY

If you find more than four, congratulations — you found something undocumented.

Alternatives for every task — every tool is replaceable

- ▶ **Diagramming** — draw.io · yEd · D2 · Mermaid · PlantUML · Excalidraw
- ▶ **Simulation** — GNS3 · EVE-NG · Cisco Packet Tracer · Containerlab
- ▶ **Scanning** — Nmap · Masscan · Zmap · Angry IP Scanner
- ▶ **Inventory** — NetBox · phpIPAM · i-doit · Snipe-IT · OCS · Open-Audit
- ▶ **Threat modelling** — ThreatDragon (OWASP) · MS Threat Modelling Tool

KEY TAKEAWAY

The competency is the eye, not the brand of the magnifying glass.

Dates · deadlines · one safety habit

- ▶ **Lab 02 due** — Sunday, 24 May 2026 · 23 h 59 EST · OmniVox LEA Dropbox
- ▶ **Late penalty** — -10 % per calendar day
- ▶ **Mid-Term Exam** — Day 07 — covers Days 01–06
- ▶ **Course Procedure Manual** — Day 14 — capstone deliverable
- ▶ **Safety habit** — verify external URLs through VirusTotal before clicking

KEY TAKEAWAY

The cheapest moment to stop is the one before you start.

Next time — Network monitoring fundamentals

- ▶ Logging architecture · what you collect · where you ship it · how long you keep it
- ▶ Wireshark + SNMP basics · setting up the lab for Lab 03
- ▶ Bring your Day 03 & 04 maps — they are the prerequisite
- ▶ Pre-read the Lab 03 brief on OmniVox before class

KEY TAKEAWAY

Day 05 lays the foundation for Lab 03 — the maps you built today are the canvas.

Legal scope · IP · safety — read once, internalise

- ▶ **IP** — © Cyber.SoHo Educational Hub — sole creator and owner · no third-party affiliation
- ▶ **URL safety** — verify external URLs through VirusTotal before clicking
- ▶ **Legal scope** — lawful ONLY on systems you own · are authorised to test · or in the lab
- ▶ **Statutes** — Canadian CC s.342.1 · US CFAA 18 USC §1030 · EU Dir. 2013/40/EU · ES LO 10/1995
- ▶ **Liability** — Cyber.SoHo disclaims all liability. If in doubt — do not perform the action.



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